

Yun Jong (Kevin) Na

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EDUCATION

University of Michigan in Ann Arbor

GPA: 3.72/4.0

B.S in Computer Science (May 2027)

Relevant Coursework: Embedded Systems, Computer Vision, Autonomous Robotics, Machine Learning, Computer Graphics, Generative Models, Computer Organization, Linear Algebra

Languages: C, C++, Python, C#; Embedded: STM32, MSP430, I²C, UART, DMA, DFSDM, ADC, PDM; Systems/Tools: ROS2, Linux (Ubuntu), NVIDIA Jetson, Intel RealSense, Git, CMake; Hardware: Oscilloscopes, logic analyzers

EXPERIENCES

Atombot Swarm Robotics Researcher, Extreme Robotics Lab

January 2025 – Present

- Co-leading a 5 member task team, developing a multi-camera AprilTag motion capture system to estimate Cookiebot poses for repeatable swarm experiments.
- Building a fast, agile, obstacle-climbing wheel-legged quadruped with a real-time sensor-fusion stack on NVIDIA Jetson (Ubuntu 20.04) + ROS2 Foxy under Prof. Y Z as part of the Multidisciplinary Program.
- Developed and validated a RealSense tracking pipeline by constructing an AprilTag ground-truth grid and benchmarking pose accuracy using mean absolute error (MAE).
- Engaged with the simulations team, learning and leveraging ROS2 and Gazebo to develop simulation frameworks for autonomous navigation and obstacle detection for wheeled quadruped robot development.

Research Assistant, Dynamic Project Management (DPM) Laboratory

Nov 2022 - 2023

- Research assistant for a Post-Doctoral Research Fellow and PhD student working in DPM Lab under the guidance of Dr. SangHyun Lee (U-M Civil & Environmental Engineering Dept).
- Labeled large-scale training datasets for machine learning algorithm from the videos collected from construction sites into different classes, like fall-related risk or not, timestamp of strides, EDA signals, and IMU signals.
- Researched on wearable based fatigue monitoring for construction workers and disadvantaged individuals, have weekly meetings and reports to the research team to communicate and provide assistance for sponsor reports.

PROJECTS

HydroSense, Tap-Triggered Water-Level Classifier (EECS 373: Embedded Systems Course)

- Built an embedded sensing pipeline on STM32L4R5ZI that classifies water level in non-transparent bottles: IMU single-tap interrupt (LSM6DSO32 via I²C) triggers PDM microphone capture, converted to 16-bit PCM at 44.1 kHz through DFSDM with double-buffered DMA, and streamed to a host over UART at 921.6k baud.
- Wrote bare-metal C firmware for I²C IMU configuration, EXTI tap-interrupt handling, DFSDM filter setup for PDM-to-PCM conversion, and DMA memory-to-peripheral transfers for WS2812B NeoPixel LED output.
- Built the Python classification pipeline: extracted MFCC features from recorded audio with librosa and trained a Linear SVM to classify six water-level classes, with predictions sent back to the MCU over UART to drive LED feedback.
- Debugged DMA timing and DFSDM oversampling configuration through iterative bench testing when documentation was insufficient, resolving 32-bit word-width and filter-ratio issues to achieve accurate PDM-to-PCM conversion.

AmbientMNIST Battery-Free Edge ML on MSP430

- Deployed a quantized CNN (5-bit fixed-point) for MNIST digit classification on an MSP430FR5994 microcontroller powered solely by ambient light via solar energy harvesting.
- Implemented intermittent-power-robust inference using the SONIC framework with dual-buffer feature maps and FRAM checkpointing, enabling computation to resume across power losses.
- Built the full deployment pipeline: trained a lightweight 2-conv + 2-FC network in Python, serialized quantized weights into C header files, and validated output via UART serial interface.

MBot Autonomous SLAM System (EECS 467: Autonomous Robotics)

- Implemented a full autonomous navigation stack in C++ on ROS2: Monte Carlo Localization (particle filter), occupancy-grid mapping with log-odds updates from LiDAR, A* path planning, and frontier-based exploration with BFS clustering.
- Developed PID motion control for differential-drive waypoint tracking and integrated all modules through ROS2 topics to enable autonomous exploration of unknown environments on physical hardware.
- Built an obstacle distance grid from the occupancy map to enforce safety constraints during path planning and frontier goal selection, filtering unreachable or unsafe targets before A* execution.

PUBLICATION

Liminal Training: Characterizing and Mitigating Subliminal Learning in Large Language Models — (NeurIPS 2025) Workshop on Socially Responsible and Trustworthy Foundation Models (ResponsibleFM), 2025.